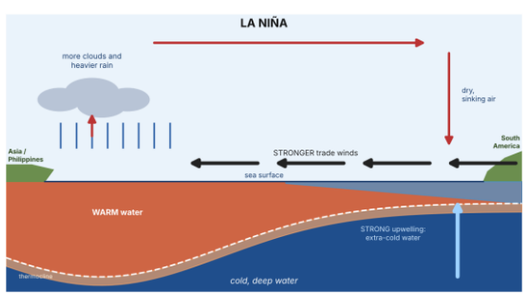
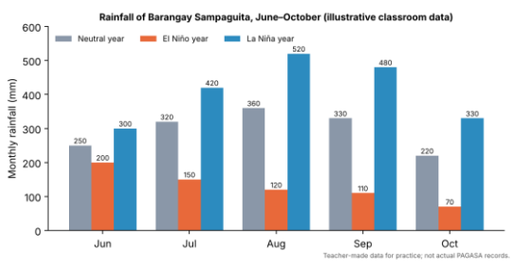
DAILY LESSON PLAN	School	**Taguig National High School**	Grade Level	**10**
	Teacher	**Ian B. Corocoto**	Learning Area	**SCIENCE (Earth Science)**
	Teaching Date and Time	**Day 3** **45 minutes**	Quarter	**Second Term (Week 9)**

I. OBJECTIVES	
A. Content Standards	Climate change and its impacts on the environment and people pose serious challenges which require solutions and action at local and global levels.
B. Performance Standards	By the end of the term, learners gather information from secondary sources to describe rapid changes that are occurring in local and global climate patterns and propose solutions to address these changes at the local and global levels by drawing on awareness, responsible personal behavior to conserve materials and energy, and through the better societal management of the natural resources of the country.
C. Most Essential Learning Competencies/ Objectives	At the end of the teaching-learning process, 100% of the students with at least 80% proficiency are expected to: 1. describe how La Niña changes the trade winds, sea-surface temperature and rainfall of the Pacific, and how these changes may impact the weather systems of the Philippines.
II. CONTENT	**Global Interactions: La Niña – The Cool Phase of ENSO and Comparison with El Niño**
III. LEARNING RESOURCES	
A. References	DepEd Grade 10 Science Budget of Work, Second Term, Week 9 – Global Interactions: "Describe how global climatic phenomena, such as the El Niño Southern Oscillation, may impact weather systems." PAGASA (https://www.pagasa.dost.gov.ph) – ENSO Alert and Warning System; La Niña advisories NOAA Climate.gov – "El Niño & La Niña" (https://www.climate.gov/enso) SunStar Cebu – PAGASA definitions of dry condition, dry spell and drought (https://www.sunstar.com.ph/cebu/pagasa-cebu-under-dry-spellnot-drought) Materials: laptop and projector, diagrams (Normal, El Niño, La Niña), manila paper, colored pencils, cut-outs (arrows, clouds, sun), rainfall graph handout, activity sheets

IV. PROCEDURES	**Teacher's Activity**	**Student's Activity**
	Classroom Conditioning (Let the teacher count 1, 2, 3)	(As the teacher counts, the students will: 1- arrange the chairs, 2- pick up the pieces of paper, and 3- be ready to pray)
	Prayer Let us pray.	(The students will pray.)
	Greetings Good morning, class! Please be seated.	Good morning, Sir! It's nice to see you. Thank you.
	Checking of Attendance Who is absent for today?	The class secretary will report the attendance for today.
A. Reviewing previous lesson or presenting the new lesson (ELICIT) *4 minutes*	Let us review yesterday's lesson using **"TRUE or FALSE – Fix It!"** If the statement is true, say "True." If false, say "False" and correct the underlined word. 1. During El Niño, the trade winds become **stronger**. 2. During El Niño, the Philippines usually receives **less** rainfall. 3. During El Niño, the warm water moves toward the **western** Pacific. 4. Three consecutive months of below-normal rainfall is called a **dry spell**.	1. False – weaker. 2. True. 3. False – the central and eastern Pacific. 4. True.
	Very good! Now, for your assignment, I asked you	(Two or three students show their

	to **predict** what happens when the trade winds become **stronger** than normal. Who would like to show their drawing?	drawings.) For example: "The warm water piles up even more in the west and cold water rises more in the east."
B. Establishing a purpose for the lesson (ENGAGE) *3 minutes*	Many of your predictions are close! Think about a year when it rained for days and many streets in your area were flooded. What do you remember about that time?	Classes were suspended and there were floods in our barangay. (Answers may vary)
	Thank you. Is it possible that the Pacific Ocean, thousands of kilometers away, played a part? Today's objective is to: **describe how La Niña changes the trade winds, sea-surface temperature and rainfall of the Pacific, and how these changes may impact the weather systems of the Philippines.**	(The students read the objective.)
C. Presenting examples/ instances of the new lesson (EXPLORE) *8 minutes*	**ACTIVITY: "BUILD THE LA NIÑA"** (Group of 5; 6 minutes) **Objective:** To use what you know about the trade winds and the warm pool to build a model of the Pacific during La Niña. **Materials:** manila paper, colored pencils, cut-outs (arrows, rain clouds, sun) **Mechanics:** 1. Draw the Pacific Ocean as a long rectangle. Put Asia/Philippines on the left and South America on the right. 2. Use the **Normal** diagram as your guide. Show what changes when the trade winds are **stronger** than normal: the arrows, the warm and cold water, and where rain clouds are. 3. Write one sentence under your drawing: "During La Niña, the Philippines will probably be ____." 4. Post your work on the board.	(The students work in groups.)
	(The teacher moves around.) Guide questions: • Where will the warm water pile up if the wind pushes harder? • What happens to the upwelling of cold water near South America?	The warm water piles up even more in the west. The upwelling gets stronger, so the sea there becomes colder.
	Time is up. Let us do a **gallery walk.** Look at the other groups' drawings and compare them with yours.	(The students walk around and observe.)
D. Discussing new concepts and practicing new skills #1 (EXPLAIN) *9 minutes*	Let us compare your models with the real La Niña diagram.  What is different from the Normal diagram?	The trade winds are stronger, the warm pool is bigger, and the cold upwelling in the east is stronger.
	Correct! During **La Niña**, the sea-surface temperature of the central-eastern Pacific becomes **cooler than normal** (scientists use about 0.5 °C or more below average, lasting several months). The strong winds pile up even more warm water in the west. What do you think happens to the rain over the	It increases, because there is more warm water and more rising air.

	Philippines?	
	Exactly! La Niña may bring **above-normal rainfall** to many parts of the Philippines, especially from late in the year to early the next year. Possible effects: • **floods and landslides** in flood-prone and mountainous areas, • **more tropical cyclones** may affect the country in some seasons, • **cooler nights** in some areas, • **crop damage** from flooding, but **more water** for irrigation and dams, • diseases that spread after heavy rain, such as **leptospirosis** and **dengue**, may increase.	(The students take notes.) Sir, so La Niña has good and bad effects.
	Great observation! That is why we say "may." The effects depend on the place and the season, and every event is different. Now let us compare the two phases of ENSO.	(The students look at the board.)
	COMPARISON (copy in your notebook): • **Trade winds** – El Niño: weaker | La Niña: stronger • **Central/eastern Pacific sea surface** – El Niño: warmer than normal | La Niña: cooler than normal • **Rain over the Philippines** – El Niño: usually below normal | La Niña: usually above normal • **Possible effects** – El Niño: dry spell, drought, heat, water shortage | La Niña: floods, landslides, more cyclones in some seasons • **PAGASA alerts** – El Niño Watch/Alert/Advisory | La Niña Watch/Alert/Advisory	(The students copy the comparison.)
E. Developing mastery (leads to Formative Assessment) (EXPLAIN) *2 minutes*	ENSO is a **cycle**: neutral → El Niño or La Niña → neutral again. An event usually lasts several months to about a year, and they do not follow a fixed schedule. Scientists call this "oscillation." Can El Niño and La Niña happen at the same time in the same place in the Pacific?	No, Sir. They are opposite phases, so one happens at a time.
F. Finding practical application of concepts and skills in daily living (ELABORATE) *6 minutes*	**ACTIVITY: "RAINFALL DETECTIVES"** (Pairs; 5 minutes) Study the graph of rainfall in a make-believe barangay (teacher-made data for practice). Then answer:  1. In which year was the rainfall highest in August?	The La Niña year (520 mm).
	2. In the El Niño year, how many consecutive months (August to October) were **more than 60% below** the neutral year? Use the PAGASA definitions. (Hint: August 360 → 120.)	Three: August, September and October. So it is a **drought**, Sir.
	3. Compute the five-month total for each year and the percent change from the neutral year. (Neutral = 1,480 mm)	El Niño: 650 mm, about 56% less. La Niña: 2,050 mm, about 39% more.
	4. If you were the barangay captain, what would you prepare for in the El Niño year and in the La Niña year?	El Niño: store water, conserve water, protect crops from heat. La Niña: clean canals, prepare evacuation centers, watch for floods and landslides.
G. Making generalizations and abstractions about the lesson (ELABORATE) *2 minutes*	Complete the statements: 1. When the trade winds are stronger than normal, the Pacific is in a _____ phase. 2. El Niño makes the Philippines usually _____, while La Niña makes it usually _____.	1. La Niña. 2. Drier; wetter. 3. Changes in the Pacific shift the rain area and can change our rainfall and temperature.

	3. ENSO affects the Philippines because _____.	
H. Evaluating learning **(EVALUATE)** *6 minutes*	**MULTIPLE CHOICE:** Write the letter of the correct answer on a ¼ sheet of paper. 1. During La Niña, the trade winds are: a) weaker than normal b) stronger than normal c) absent d) blowing west to east 2. During La Niña, the sea surface of the central and eastern Pacific is: a) warmer than normal b) frozen c) cooler than normal d) exactly the same 3. Which is a likely effect of La Niña on many parts of the Philippines? a) Above-normal rainfall and flood risk b) Long drought for a year c) No clouds d) Snow	1. B 2. C 3. A
	4. Which pair is correct? a) El Niño – wetter Philippines; La Niña – drier Philippines b) El Niño – drier Philippines; La Niña – wetter Philippines c) Both always bring typhoons d) Both have the same effect 5. Based on the graph, which statement is supported by the data? a) The El Niño year had about 56% less rain than the neutral year, and the La Niña year about 39% more. b) The La Niña year had less rain than the neutral year. c) The El Niño year had the highest rainfall. d) All three years had about the same rainfall.	4. B 5. A
I. Additional activities for application or remediation **(EXTEND)** *1 minute*	**Assignment (for the Day 4 activity):** 1. Search for one Philippine news story (from PAGASA, DOST, DA, or a national newspaper) about a drought, dry spell, flood, or heat advisory. Write the date, the place, and the effect on people. 2. Review El Niño, La Niña, and the PAGASA terms. There will be a short quiz and a group activity tomorrow. Thank you, class! Goodbye.	(The students write the assignment in their notebooks.) Goodbye and thank you, Sir!

J. REFLECTION	
A. No. of learners who earned at least 80% in the evaluation	
B. No. of learners who require additional activities for remediation	
C. Did the remedial lessons work? No. of learners who have caught up with the lesson	
D. No. of learners who continue to require remediation	
E. Which of my teaching strategies worked well? Why did these work?	

F. What difficulties did I encounter which my principal or supervisor can help me solve?
G. What innovation or localized materials did I use/discover which I wish to share with other teachers?

Prepared by:

IAN B. COROCOTO
Student Teacher

Checked:

[COOPERATING TEACHER]
Critic Teacher
Master Teacher 1

Noted:

[HEAD TEACHER]
Head Teacher III

Name: _____ Group: _____ Score: _____

"RAINFALL DETECTIVES" – Rainfall of Barangay Sampaguita (illustrative classroom data)

Month	Neutral year (mm)	El Niño year (mm)	% change	La Niña year (mm)	% change
June	250	200		300	
July	320	150		420	
August	360	120		520	
September	330	110		480	
October	220	70		330	
TOTAL	1,480				

Formula: % change = (new value − neutral value) ÷ neutral value × 100

Guide Questions:

1. In which year was August rainfall highest?
2. In the El Niño year, how many consecutive months were more than 60% below the neutral year? What does PAGASA call this?
3. What are the five-month totals and percent changes?
4. What should the barangay prepare for in each year?

Teacher's key: % change (El Niño): −20, −53, −67, −67, −68; total 650 mm (≈ −56%). % change (La Niña): +20, +31, +44, +45, +50; total 2,050 mm (≈ +39%). Aug–Oct are more than 60% below normal for three consecutive months, which meets PAGASA's description of drought.

Graphic Organizer – Compare and Contrast

Feature	El Niño	Both	La Niña
Trade winds			
Central/eastern Pacific sea surface			
Rain over the Philippines			
Possible effects			

Rubric for "Build the La Niña" (20 points)

Criteria	4	3	2	1
Scientific accuracy of the model	All parts correct	Mostly correct	Some errors	Mostly wrong
Labels and neatness	Complete and neat	Mostly complete	Incomplete	None
Prediction sentence about the Philippines	Correct and explained	Correct	Partly correct	Missing
Teamwork	Everyone participated	Most participated	Few participated	One member
Time	Finished early/on time	On time	Late	Not finished

Answer Key – Day 3 Quiz

1. B 2. C 3. A 4. B 5. A